

ANALOG CIRCUITS – CIRCUIT DEBUGGING ASSESSMENT TOOL

Q11. Debugging of BJT Amplifier Using LTspice

Circuit Debugging of a BJT Amplifier Using LTspice

Circuit debugging is the systematic process of locating why a simulated circuit does not behave as intended, by tracing the signal path node-by-node and checking each component and bias condition against the expected topology. LTspice **transient analysis** (.tran) displays time-domain voltages at every node, making it possible to identify exactly where a signal is lost. Here, V_{in} is present but the transient output across R_L is zero. This report identifies the possible causes, explains the underlying theory, and verifies the corrected circuit through simulation.

1. Initial Circuit Inspection

The supplied circuit (Fig. 1) has an NPN BJT, $R_C = 4\text{ k}\Omega$, an emitter resistor R_E (value not given), an output coupling capacitor $C_2 = 10\text{ }\mu\text{F}$, a load $R_L = 2\text{ k}\Omega$, a supply V_{CC} , and a sinusoidal source V_{in} at the base.

Topology observation: the circuit is topologically a **Common-Emitter (CE) amplifier**, not a Common-Collector (Emitter Follower), because the output is taken from the collector through C_2 . In a true emitter follower the collector ties directly to V_{CC} (no R_C) and the output is taken from the *emitter* node across R_E . Since R_C sits between V_{CC} and the collector, and the output is coupled out of the collector, the circuit must be analysed as a CE amplifier throughout.

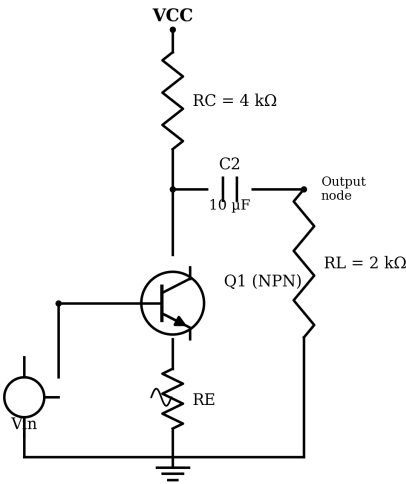


Figure 1. Circuit as supplied in the question. Output is taken from the collector through C_2 — this is a Common-Emitter (CE) topology, not an Emitter Follower (which would take output from the emitter node).

Figure 1. Circuit as supplied: output taken from the collector via C_2 — a Common-Emitter amplifier, not an Emitter Follower.

2. Expected Signal Path

$V_{in} \rightarrow \text{Base} \rightarrow \text{Transistor (amplified, inverted)} \rightarrow \text{Collector} \rightarrow C_2 \rightarrow R_L \rightarrow \text{Ground}$

The base signal modulates collector current, producing an amplified, inverted AC voltage at the collector. This AC component must pass through C_2 (which blocks the DC bias) before it can appear across R_L . A break anywhere along this path — base, transistor, C_2 , or the R_L /ground return — gives zero measured output even though V_{in} is healthy.

3. Possible Faults

Items (a)–(d) are the four faults raised in the question and are the primary suspects; (e)–(g) are additional faults a thorough procedure must also rule out.

Fault	Effect	LTspice Observation	Correction
a) Wrong transistor pins (B/C/E)	Device forced into cutoff, saturation, or reverse-active; normal amplification lost.	Collector stays at a fixed DC rail value; no AC synchronized with V_{in} .	Verify B-C-E against the datasheet pinout; correct wiring/orientation.
b) Open / faulty C_2	No AC path exists between collector and R_L ; DC-blocking also removes the only AC route.	AC visible at collector node, but $V(R_L)$ trace is flat at 0 V.	Replace with a functioning $10\text{ }\mu\text{F}$ capacitor; check continuity.

Fault	Effect	LTspice Observation	Correction
c) Missing VCC	No collector current; transistor has no Q-point and cannot amplify.	Collector sits near 0 V; no meaningful signal past the base.	Confirm VCC source present, nonzero, wired to top of RC.
d) Improper / absent grounding	LTspice needs one common reference (node 0); return paths for AC current break without it.	Simulation fails to converge, or voltages are meaningless/floating.	Tie source return, emitter network, and RL return to one ground.
e) Wrong transistor model / orientation	Wrong SPICE model (e.g. PNP) or mirrored symbol reverses current direction and bias polarity.	Transistor appears permanently off, or currents flow backward.	Select correct NPN model (e.g. 2N2222); confirm symbol orientation.
f) Improper biasing / cutoff	No DC base bias base stays below the 0.7 V turn-on threshold for most of the cycle.	Collector waveform heavily clipped, distorted, or absent.	Add bias network (R1/R2 divider) so Q-point lies in active region.
g) Incorrect C2–RL wiring	Open node or wrong net prevents signal reaching load even with a healthy C2.	V(RL)=0 despite healthy AC at the C2/collector-side node.	Verify C2's right plate ties to RL top; RL's other end to ground.

Technical Analysis of the Fault Conditions

1. Incorrect Transistor Pin Connections

A BJT's behaviour depends entirely on which lead performs which role. Interchanging collector and emitter, or tying the base to the wrong net, can force reverse-active operation (very low gain), cutoff ($I_C \approx 0$), or saturation ($V_{CE} \approx 0$). In LTspice, an incorrectly rotated/mirrored transistor symbol places the wrong physical terminal at each net without raising an error — it simply produces incorrect operation, so the collector no longer carries the expected AC signal regardless of C2's condition. Pin configuration must be verified against the specific datasheet (e.g. 2N2222: E-B-C physical pinout in TO-92).

2. Open Coupling Capacitor C2

C2 is the *only* electrical path between the collector node and RL. A capacitor's opposition to AC is its reactance:

$$X_C = 1 / (2\pi f C)$$

For a healthy capacitor this is small at signal frequencies. Example at $f = 1$ kHz, $C2 = 10$ μ F:

$$X_C = 1 / (2\pi \times 1000 \times 10 \times 10^{-6}) \approx 15.9 \, \Omega$$

This is small next to $RC = 4$ k Ω and $RL = 2$ k Ω , so a healthy C2 behaves nearly as a short for AC. If C2 is **open**, its impedance is effectively infinite at all frequencies: the collector AC signal cannot reach RL, so $V(RL) \approx 0$ even though the collector itself may still show a healthy amplified waveform. This is the single fault that most directly and exclusively explains “Vin present, $V_{RL} = 0$ ” in an otherwise correctly biased amplifier.

3. Missing Supply Voltage (VCC)

$$V_C = V_{CC} - I_C R_C$$

Without VCC, no collector current flows through RC, so the transistor cannot be biased into the active region and cannot amplify. The collector then stays at a fixed low potential regardless of the input, so no AC component is available to couple through C2 to RL.

4. Improper Grounding

LTspice solves node voltages relative to one reference node (node 0). The source return, emitter network (through RE), and load return (through RL) must all share this reference. A missing ground on any branch can stop convergence or make the computed voltages meaningless, since they are not referenced to the same point as the rest of the circuit — so RL can show no meaningful signal even if the amplifier itself is otherwise correct.

5. Biasing Problem (Transistor in Cutoff)

A silicon BJT needs ≈ 0.7 V of forward bias across base-emitter before conducting. With a pure AC Vin and no bias network, the base-emitter junction stays below turn-on for most or all of the cycle, so the transistor remains in cutoff ($I_C \approx 0$, $I_E \approx I_C \approx 0$), giving a clipped or absent output. A defined Q-point — e.g. via a voltage-divider (R1, R2) — is required so Q1 stays active for the full input swing, using $I_E \approx I_C$ and $V_E = I_E R_E$ to set the operating current.

LTspice Debugging Procedure and Corrected Circuit

The following sequential procedure isolates each fault using standard LTspice checks (node probes, .op analysis, and .tran transient analysis).

STEP 1 – Ground. Confirm a ground symbol (node 0) exists and the source return, emitter network, and load return all connect to it; run .op first — a floating node is reported immediately.

STEP 2 – VCC. Confirm a nonzero VCC source is present. **VCC = 12 V** is used here as an assumed simulation value, since the question does not specify a numerical supply.

STEP 3 – Transistor pins. Verify B, C, E against the datasheet. An NPN **2N2222** is used for demonstration and can be replaced by the lab-specified model without changing the method.

STEP 4 – C2. Confirm $C2 = 10\ \mu\text{F}$ and not open; replace with a functioning part and check polarity if a polarized model is used.

STEP 5 – RL. Confirm $RL = 2\ \text{k}\Omega$ is connected from the node after C2 to ground, not directly to the collector (which would bypass C2).

STEP 6 – Biasing. Add a base bias network so Q1 stays active for the full input swing. Assumed values used below are explicitly labelled and are not given in the question:

Parameter	Assumed Simulation Value
VCC	12 V (assumed)
RC	4 k Ω (given)
RE	1 k Ω (assumed — not given)
C2	10 μF (given)
RL	2 k Ω (given)
Transistor	2N2222 NPN (assumed)
Input amplitude / frequency	10 mV / 1 kHz (assumed)

Table: assumed values for demonstration only — clearly distinguished from the given values (RC, C2, RL).

With these values, VCC feeds RC to the collector; the base gets a stable DC bias from R1/R2 with Vin coupled in through Cin; the emitter is stabilised through RE; and the collector AC signal is coupled to RL through a verified, functioning C2 — all referenced to one common ground.

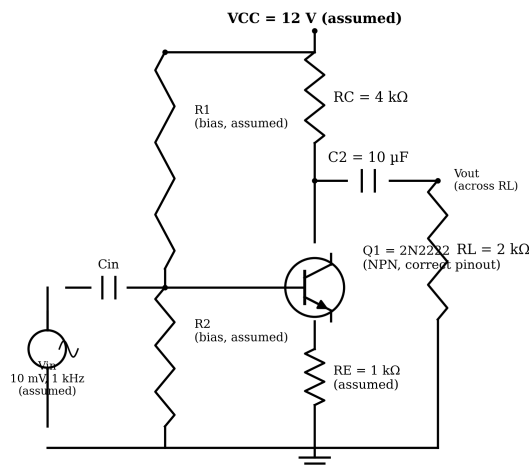


Figure 2. Corrected circuit: verified transistor pin connections, functioning C2, VCC applied, common ground established, and a voltage-divider bias network (R1, R2) added so Q1 operates in the active region.

Figure 2. Corrected circuit: verified pin connections, functioning C2, applied VCC, common ground, and voltage-divider bias (R1, R2) so Q1 operates in the active region.

Fault	Before Correction	After Correction
Open C2	RL output $\approx 0\ \text{V AC}$	AC signal appears across RL
Missing VCC	No transistor operation	Amplified collector signal
Wrong transistor pins	Incorrect / no output	Correct transistor operation
No ground	Invalid simulation reference	Stable, referenced waveform

Fault	Before Correction	After Correction
Improper bias	Cutoff / distorted output	Proper amplified waveform

Transient Analysis and Verification After Debugging

After correction, the circuit is re-simulated with:

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.tran 0 10m
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At 1 kHz (period = 1 ms), a 10 ms window captures ≈ 10 cycles — enough to observe steady-state amplitude and phase after any initial transient settles. **Before debugging:** V_{in} is present at the base, but $V(RL) \approx 0$ whenever the path is interrupted (open C2, missing VCC, wrong pins, or broken ground) — even though V_{in} itself is healthy. **After debugging:** the collector carries a DC bias plus an amplified AC component, which passes through the healthy C2 to appear across RL. For a CE amplifier in its midband region, the output is expected to be approximately **180° out of phase** with the input.

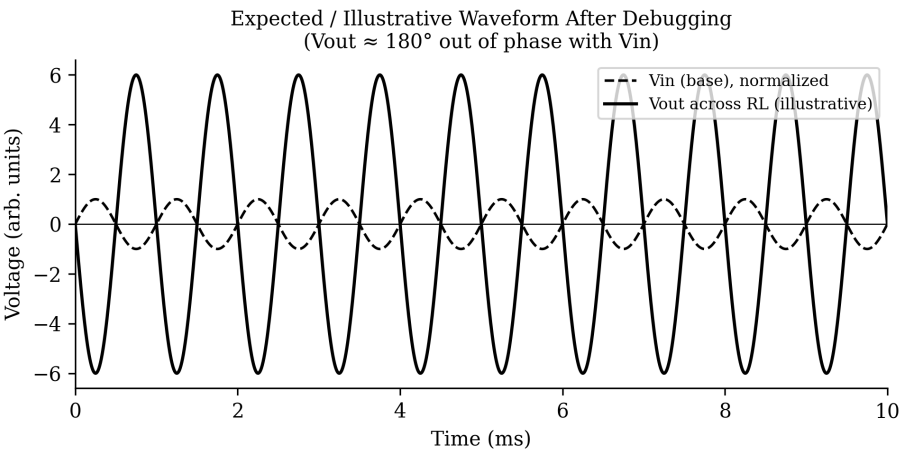


Figure 3. Expected/illustrative waveform after debugging. Exact amplitude is not claimed (it depends on h_{FE} and bias point); the figure shows the expected 180° phase inversion, not simulated amplitude data.

Gain is expressed qualitatively as $A_v = V_o/V_i$; its exact value depends on transistor parameters, bias point, and loading. Because RL is coupled through C2, the effective AC collector load in midband is the parallel combination of RC and RL, not RC alone.

Verification Checklist

- VCC present

Ground connected

Correct transistor pin configuration

Correct transistor model

C2 = 10 μ F and not open
- RL = 2 k Ω correctly connected

Proper transistor biasing

Input signal present

Output measured across RL

Transient simulation completed successfully

Conclusion

A zero output across RL despite a healthy V_{in} does not mean the input source is faulty — it shows that the AC signal path or the transistor's DC operating conditions are compromised. For an otherwise correctly wired amplifier, the single most direct explanation is an **open coupling capacitor C2**, since it is the only AC path from collector to RL: if open, the collector may still show a healthy signal while RL reads zero. Other causes are equally capable of producing this symptom: incorrect transistor pin connections place the device outside its active region; a missing VCC removes the DC operating point entirely; improper or absent grounding invalidates the simulation reference; incorrect C2–RL wiring breaks continuity even with a healthy capacitor; and inadequate base biasing can leave the transistor in cutoff for most of the cycle. A systematic procedure — checking ground, supply, pinout, coupling capacitor, load wiring, and bias point in turn — isolates the true cause efficiently. Once corrected, LTspice transient analysis confirms a proper amplified, phase-inverted waveform across RL, consistent with expected Common-Emitter behaviour.